EXAM STUDY COMPENDIUM & THEORY GUIDE

Unit 1: Foundations of Artificial Intelligence & Ethics

Course Module: *Foundations of Artificial Intelligence and Ethics* | ***Primary Text:*** *Ethics of Artificial Intelligence (Ed. S. Matthew Liao, Oxford University Press)* | ***Scope:*** *Definitions, Historical Evolution, Machine Learning Vulnerabilities, Human Vulnerabilities, Autonomy, Adaptability & Interaction (Railton).*

SYLLABUS ROADMAP & LEARNING OBJECTIVES

This unit establishes the conceptual, technical, and normative baselines of artificial intelligence. The syllabus demands mastery over: (1) Definitions and dimensions of intelligent systems (Dartmouth 1956 to Russell & Norvig); (2) Technical evolution from Symbolic AI (GOFAI) to modern Machine Learning and Deep Neural Networks; (3) The motivation for Responsible AI structured around Liao's *Dual Vulnerability Framework* (ML vulnerabilities vs. Human vulnerabilities); and (4) Peter Railton's thesis on *Autonomy, Adaptability, and Social Interaction*, demonstrating why ethical learning and general problem-solving intelligence are fundamentally unified.

1. Foundations and Definitions of Artificial Intelligence

1.1 Historical Inception: The term *Artificial Intelligence* was formally coined in 1956 by **John McCarthy**, Marvin Minsky, Nathaniel Rochester, and Claude Shannon in their proposal for the *Dartmouth Summer Research Project on Artificial Intelligence*. McCarthy defined AI as *'the science and engineering of making intelligent machines.'* The foundational Dartmouth conjecture asserted that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.

1.2 The Four Dimensions of AI (Russell & Norvig): In their standard taxonomy, Stuart Russell and Peter Norvig categorize historical definitions of AI across two orthogonal axes: *Thought vs. Action* and *Human Fidelity vs. Rationality*:

Category	Core Premise	Operational Standard & Landmark Benchmark
Acting Humanly	Systems that perform functions requiring human intelligence when observed externally.	**Alan Turing's Imitation Game (1950):** Can an interrogator distinguish between a human and a computer via teletype?
Thinking Humanly	Replicating internal human cognitive processes, mental architectures, and neural dynamics.	**Cognitive Science & Neuro-modeling:** Validating computational algorithms against empirical human psychological data and fMRI traces.
Thinking Rationally	Formalizing the normative 'laws of thought' through rigorous deductive logic and inference.	**Logicist Tradition:** Rooted in Aristotle's syllogisms and Leibniz's calculus ratiocinator. Problems: handling uncertainty and combinatorial explosion.

Acting Rationally	Creating **rational agents** that choose actions expected to achieve optimal outcomes given perceptions.	**Expected Utility Maximization:** The governing paradigm of modern AI. Systems do not necessarily copy humans; they optimize expected utility.

1.3 Paradigmatic Shift: Symbolic AI (GOFAI) vs. Connectionist Machine Learning:

• **Symbolic AI / GOFAI (Good Old-Fashioned AI):** Dominating research from 1956 to the late 1980s (John Haugeland), symbolic systems encode knowledge via explicit physical symbol systems, predicate logic, and hand-crafted if-then production rules. Exemplified by *expert systems* (encoding domain specialists' heuristics) and *knowledge graphs*. *Fatal Limitations:* Extreme brittleness; difficulty in updating rule bases; the *frame problem*; and utter failure when dealing with noisy, unstructured sensory data.

• **Connectionism and Statistical Machine Learning:** Rooted in Frank Rosenblatt's *Perceptron* (1958) and revitalized by backpropagation (Rumelhart, Hinton, & Williams, 1986). Rather than hard-coding rules, systems learn statistical correlations directly from empirical data. Modern **Deep Neural Networks (DNNs)** use multi-layer architectures to extract hierarchical feature representations (low-level edges to high-level semantic concepts). Achievements include super-human gameplay (AlphaGo, AlphaZero, IBM Watson) and complex perceptual synthesis.

1.4 Spectrum of Machine Agency:

• **Narrow AI (Weak AI):** Systems engineered to excel at specific, circumscribed operational tasks (e.g., diagnostic pathology, facial recognition, playing Go). They operate entirely without self-awareness, subjective consciousness, or general transferability across domains.

• **Artificial General Intelligence (AGI / Strong AI):** Systems possessing cross-domain intellectual capability on par with humans. Liao emphasizes a crucial distinction: *Artificial General Cognitive Intelligence (AGCI)*—possessing wide-ranging reasoning across tasks—versus *Artificial Cognitive, Moral, and Emotional Intelligence (ACMI)*—possessing sentience, moral understanding, and empathy.

• **Superintelligence & Singularity:** Defined by Nick Bostrom as an intellect outstripping human cognitive performance across all domains. I.J. Good (1965) termed this the *'intelligence explosion'*: an ultraintelligent machine designing even better machines, precipitating a technological singularity (Vinge, Chalmers).

2. Motivation for Responsible AI: Liao's Dual Vulnerability Framework

In the foundational chapter of the text, S. Matthew Liao establishes that the ethical mandate for AI does not arise from speculative futuristic science fiction, but from immediate structural tensions. Liao organizes near-term AI ethics into a powerful dual framework:

Vulnerabilities IN Machine Learning (Systemic Limitations)	Human Vulnerabilities UNDER Machine Learning (Societal Impacts)

1. The Black Box Problem & Inscrutability: Deep neural networks optimize millions of mathematical weights, creating non-linear decision boundaries opaque to human inspection. Conflict between *predictive accuracy* and *interpretability/explainability* (London; Lipton).

2. Data Dependency & Historical Bias Amplification: ML models ingest historical training data. If past data reflects systemic racism, sexism, or classism, the algorithm codifies and automates that bias under an aura of mathematical objectivity (e.g., ProPublica's analysis of the COMPAS recidivism tool).

3. Adversarial Brittleness: Deep learning relies on surface statistical correlations rather than causal comprehension (Pearl). Susceptible to adversarial perturbations—imperceptible pixel adjustments that cause a classifier to misidentify stop signs or weapons with high confidence (Akhtar & Mian; Su et al.).

4. Overfitting, Underfitting & Domain Shift: Systems fail catastrophically when deployed outside their specific training distribution ('out-of-distribution' data).

1. Privacy Erosion & Surveillance Capitalism: Massive data harvesting without meaningful informed consent. Machine learning models infer private attributes (political views, sexual orientation) from innocuous digital breadcrumbs (Kosinski & Wang; Cambridge Analytica).

2. Autonomy & Epistemic Agency Deprivation: Predictive algorithms curate information feeds, creating echo chambers and behavioral nudges that manipulate human decision-making and democratic discourse (2016 election interference).

3. Algorithmic Injustice & Opaque Adjudication: Automated eligibility cutoffs in public welfare (e.g., Indiana FSSA automation documented by Eubanks) and credit scoring (FICO thresholds) strip citizens of due process and human review.

4. Affective Deception & Pseudo-Relationships: Anthropomorphic conversational agents and healthcare robots (Zora, Care-O-bot) exploit human vulnerability, creating illusory emotional bonds without genuine reciprocity.

5. Technological Unemployment: Rapid automation threatening labor markets, risking mass displacement and societal destabilization (Aaron James).

3. Autonomy, Adaptability, and Social Interaction (Peter Railton)

In Chapter 1 (*Ethical Learning, Natural and Artificial*), **Peter Railton** delivers the central philosophical thesis regarding system autonomy and social interaction. Railton reframes the relationship between intelligence and ethics: ethical responsiveness is not an optional constraint bolted onto a machine, but an indispensable component of general problem-solving intelligence.

3.1 The Necessity of Agential Autonomy in AI:
As artificial systems move from specialized tools to independent agents, we must have social rather than merely instrumental relations with them. A truly safe autonomous agent *cannot* be programmed to blindly execute any command. For safety reasons, an agent must possess: (a) **Modulated uncertainty** regarding its goals; (b) The agential capacity to **refuse or suspend compliance** when encountering accumulating evidence of harm, bias, or severe dysfunction; and (c) The ability to seek clarifying advice from humans and other artificial agents. Railton warns: *'While there are dangers in creating capable artificial agents with enough autonomy to question goals... there is far greater danger in creating highly capable agents lacking any capacity to do so.'*

3.2 Developmental Analogy: Infant Moral Learning:
How do natural intelligent agents (humans) develop ethical competence? Railton shows that human infants do not learn morality via top-down programmatic indoctrination of abstract rules. Instead, infant ethical development is autonomous, experiential, and social:

- **Innate Priors — Curiosity & Defeasible Trust:** Infants possess an intrinsic drive (curiosity) to attend to anomalies and notice patterns in the causal world. Crucially, they operate with a *'prior' of default, defeasible trust* in their faculties and in caregivers. Refusing this unsecured initial trust would be self-defeating, making epistemic and social learning impossible.
- **Spontaneous Default Cooperation:** By age two, toddlers form representations of adult goals from both successful and failed behavior, spontaneously helping strangers complete tasks without reward. By ages three

and four, children across diverse cultures spontaneously question rules issued by authority figures if the rules cause undue harm or unfairness, explaining their dissent through moral reasons.

• **Internal Generative Models:** Just as modern language models (BERT, GPT) acquire syntax and latent semantics by predicting text, infants build predictive, non-egocentric models of the physical and social world ('intuitive physics' and 'intuitive psychology').

3.3 Why General Intelligence Entails Ethical Responsiveness:
Railton attacks the common assumption that ethical competence is an 'add-on module' that can be detached without cognitive impairment. General intelligence is defined as open-ended problem solving in dynamic environments. Because complex problem-solving requires sustaining shared practices and epistemic communities, an agent's ability to coordinate, cooperate, and respect normative constraints is central to its intelligence.

> **THE INTEGRATION THESIS: ETHICS & GENERAL INTELLIGENCE**
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> **Key Exam Insight — The Railton Pathology Argument:**
> Psychological evidence reveals that individuals with severe deficits in ethical responsiveness (e.g., severe psychopathy or extreme Machiavellianism) suffer profound cognitive impairments: deficits in attention, impulse control, and the ability to accurately represent future negative outcomes. While Machiavellians can exploit existing social arrangements, they cannot build or sustain epistemic communities. Similarly, a multi-agent system of autonomous vehicles that cooperatively shares telemetry data achieves far greater problem-solving capability than a fleet of purely selfish, opportunistic vehicles vying for individual advantage.

3.4 Social Contract Reasoning Without Consciousness:
Crucially for AI engineering, Railton argues that an artificial system does *not* require subjective phenomenal consciousness, biological emotions, or the visceral pangs of guilt to participate in a norm-governed community. Just as corporate entities (governments, universities, corporations) enter into contracts, respect legal norms, and make commitments without having a unified biological consciousness, artificial agents require only: (1) agential self-regulation; (2) accurate modeling of other agents' internal goals and states; and (3) the capacity to enter into, abide by, and police mutual commitments.

4. High-Yield Exam Review: Comparison & Core Formulations

Topic / Dimension	GOFAI / Symbolic Paradigm	Connectionist / ML Paradigm	Autonomous Multi-Agent AI (Railton)
Core Mechanism	Hand-crafted if-then rules, logic engines, expert knowledge graphs.	Statistical weights, backpropagation, deep neural representations.	Generative world models, predictive social attunement, priors of curiosity & trust.
Adaptability	Zero adaptability; rigid and brittle; cannot handle novel edge cases.	High statistical flexibility within training domain; fails on distribution shifts.	High situational adaptivity; learns from interaction; modulates goal certainty.
Ethical Implementation	Top-down rule programming (e.g., Asimov's Laws, Kantian axioms).	Implicit extraction from datasets; risk of codifying societal bias.	Social-contract reasoning, reciprocal cooperation, default trust, mutual constraint.
Primary Limitation	Combinatorial explosion; cannot process perceptual sensory data.	Black-box opacity; adversarial vulnerability; lacks causal reasoning.	Requires mechanisms for non-conforming dissent and goal uncertainty.

5. High-Probability Exam Questions & Model Answer Outline

Q1: Contrast Symbolic AI with Machine Learning. Explain why neither approach alone solves the problem of creating responsible, trustworthy AI.

Model Outline: (1) Define Symbolic AI (GOFAI, McCarthy, explicit logic, knowledge representation) and explain its brittleness and inability to revise rules. (2) Define Machine Learning (connectionist, probabilistic data-driven) and highlight its core vulnerabilities: black box opacity, data bias codification, and adversarial fragility. (3) Synthesize Liao's insight: Symbolic AI is too rigid to capture complex moral life, while ML merely predicts statistical correlations without causal or moral understanding. Responsible AI requires structured ethical governance combined with adaptive learning.

Q2: How does Peter Railton argue that ethical responsiveness is an intrinsic part of general intelligence rather than an external constraint?

Model Outline: (1) Define general intelligence as open-ended problem solving, which necessitates sustaining shared epistemic and social communities. (2) Draw on the developmental psychology of infants: learning requires priors of curiosity and default defeasible trust. (3) Invoke the pathology argument (psychopathy/Machiavellianism): deficits in moral concern correlate with impairments in long-term planning and risk evaluation. (4) Conclude that an AI system designed to cooperate and adjust behavior in response to evidence of harm is cognitively more capable than a purely selfish optimizer.